

Driver Drowsiness Prediction

Ayah Said
Mohammad Al-Omaishat
Bani Meshal

Abstract—Driver drowsiness is one of the leading causes of road accidents and can significantly reduce a driver’s reaction time and decision-making ability. This project investigates the use of physiological signals to estimate driver drowsiness [8] by predicting the Karolinska Sleepiness Scale (KSS) [1]. This report focuses on the design, implementation, and evaluation of a regression Artificial Neural Network (ANN) for predicting continuous KSS values from six extracted photoplethysmography (PPG) features. Different validation strategies were compared to evaluate the model’s performance and its ability to generalize to unseen data.

Index Terms—Driver Drowsiness, Artificial Neural Network, Regression, PPG, Karolinska Sleepiness Scale

I. Introduction

Driver drowsiness is a major contributor to traffic accidents and poses a serious risk to road safety [8]. As fatigue increases, drivers experience reduced attention, slower reaction times, and impaired decision-making, which can lead to dangerous driving situations. Detecting drowsiness at an early stage can therefore improve road safety and support the development of intelligent driver assistance systems.

Physiological signals provide valuable information about the human body’s response to fatigue. Among these signals, photoplethysmography (PPG) offers a non-invasive method for monitoring cardiovascular activity and extracting meaningful physiological features [7]. Instead of using raw PPG signals directly, signal processing techniques can be applied to obtain representative features that are more suitable for machine learning models.

The objective of this project is to predict the Karolinska Sleepiness Scale (KSS) [1], a continuous measure of subjective sleepiness, using physiological features extracted from PPG recordings. Within the overall project, this report focuses on the development and evaluation of a regression Artificial Neural Network (ANN). The model was trained to estimate continuous KSS values from six extracted PPG features, and its performance was evaluated using multiple validation strategies to investigate its ability to generalize under different testing conditions.

II. Approach

A. Overall Pipeline

The overall workflow of the project is illustrated in Fig. 1. Raw photoplethysmography (PPG) recordings [7] were first preprocessed to improve signal quality. Relevant physiological features were then extracted from the processed signals and used as input to a Regression

Artificial Neural Network (ANN). The network predicts a continuous Karolinska Sleepiness Scale (KSS) score, which represents the driver’s level of drowsiness.

It should be noted that the preprocessing and feature extraction stages were completed as part of the overall team project. The contribution presented in this report begins after feature extraction and focuses on the design, implementation, training, optimization, and evaluation of the Regression ANN.

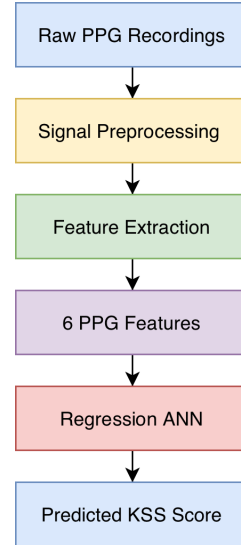


Fig. 1. Overall workflow of the proposed driver drowsiness prediction system, from raw PPG recordings to continuous KSS prediction using the regression ANN.

The dataset was collected through internal recording sessions using a static driving simulator. During each session, physiological signals were recorded with a BITalino device [6], while a face camera captured facial features. Driver drowsiness was measured using the Karolinska Sleepiness Scale (KSS) [1], which served as the ground-truth label. The recordings were divided into one-minute segments, and after preprocessing and feature extraction, six PPG-derived features were used as inputs to the Regression ANN presented in this report.

B. Data Acquisition Challenge

During the data collection phase, a channel mapping issue was encountered in the Somnus framework used to communicate with the BITalino device. Due to an incorrect channel index in the capture method, the recorded

physiological channels were shifted, causing several signals to be assigned incorrect labels.

As a result, the channel labeled as RESP contained the PPG signal, while the channels labeled as EDA and ECG corresponded to different physiological signals than expected. Furthermore, the ECG recordings could not be recovered and were therefore excluded from further analysis.

Once the correct channel mapping had been identified, the PPG signal was used throughout the preprocessing and feature extraction stages. Since the Regression ANN developed in this work relies exclusively on PPG-derived features, the corrected mapping ensured that valid input data were used for training and evaluation.

C. Feature Set

The Regression ANN operates on six features extracted from the processed PPG recordings rather than the raw signal itself. These features include the mean heart rate (PPG_Rate_Mean), heart rate standard deviation (PPG_Rate_STD), mean signal quality (PPG_Quality_Mean), peak count (Peak_Count), mean peak interval (Peak_Interval_Mean), and peak interval standard deviation (Peak_Interval_STD). Together, these features provide a compact representation of the cardiovascular signal and serve as the input vector to the Regression ANN.

The extracted features capture complementary aspects of the PPG signal. PPG_Rate_Mean and PPG_Rate_STD describe the average heart rate and its variability during each recording segment. PPG_Quality_Mean estimates the quality of the recorded signal, while Peak_Count represents the number of detected pulse peaks. Finally, Peak_Interval_Mean and Peak_Interval_STD describe the average time between consecutive heartbeats and its variability. Together, these features provide a compact physiological representation while significantly reducing the dimensionality of the original PPG signal.

D. Network Architecture

A Regression Artificial Neural Network (ANN) was developed to predict continuous Karolinska Sleepiness Scale (KSS) values from the six extracted PPG features. Since KSS is represented as a continuous numerical value, the prediction task was formulated as a regression problem rather than a classification problem [5].

The proposed network consists of an input layer receiving the six extracted PPG features, followed by two fully connected hidden layers with 16 neurons each. Both hidden layers employ the Rectified Linear Unit (ReLU) activation function [3] to capture nonlinear relationships between the physiological features and the target variable. The output layer contains a single neuron without an activation function, allowing the network to produce continuous KSS predictions.

The network architecture was intentionally designed to remain relatively compact because the available dataset contained a limited number of participants and training samples. Increasing the complexity of the model could increase the risk of overfitting, where the network memorizes the training data instead of learning general physiological patterns. Two hidden layers were therefore selected as a compromise between expressive power and model simplicity.

Dense (fully connected) layers were chosen because the input consists of a small numerical feature vector rather than images or sequential data. This architecture enables every extracted feature to contribute to the prediction while allowing the network to learn interactions between heart rate, signal quality, and peak-based measurements.

The ReLU activation function [3] was selected for the hidden layers because it is computationally efficient, accelerates network training, and effectively models nonlinear relationships without suffering from the vanishing gradient problem associated with traditional activation functions such as sigmoid or tanh. Since the objective is to estimate continuous KSS values, the output layer uses a linear activation (no activation function), allowing unrestricted numerical predictions.

The number of neurons was selected experimentally. Preliminary experiments with larger hidden layers showed no consistent improvement in prediction accuracy. Consequently, 16 neurons per hidden layer were retained because they provided a good balance between learning capacity, computational efficiency, and generalization performance.

The Regression ANN was implemented using TensorFlow/Keras [4] and trained using the Adam optimization algorithm [2] with the default learning rate. Adam was selected because it combines the advantages of adaptive learning rates and momentum, allowing stable and efficient optimization without extensive manual parameter tuning.

The network was trained for 100 epochs using separate training and validation datasets. During each epoch, the model parameters were updated in mini-batches of training samples, while the validation data were used to monitor the model's performance throughout training.

Mean Squared Error (MSE) was selected as the loss function because it is one of the standard optimization objectives for regression problems [5]. By penalizing larger prediction errors more heavily than smaller ones, MSE encourages the network to minimize substantial deviations between the predicted and true KSS values.

Although MSE was used during optimization, Mean Absolute Error (MAE) was chosen as the primary evaluation metric because it directly represents the average prediction error in KSS units, making the results easier to interpret and compare across the different validation strategies [5]. This metric also corresponds to the performance measure used throughout the experimental evaluation.

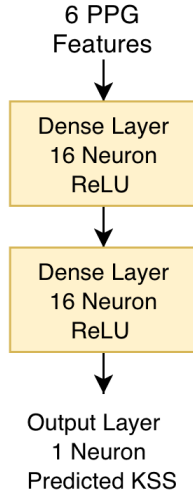


Fig. 2. Architecture of the proposed Regression Artificial Neural Network.

E. Hyperparameter Optimization

Several hyperparameter configurations were investigated to improve the predictive performance of the Regression ANN while maintaining a simple and robust architecture. The experiments focused on identifying a network configuration that achieved good prediction accuracy without introducing unnecessary model complexity or overfitting.

One of the investigated techniques was EarlyStopping, a commonly used regularization technique in deep learning [5], which automatically terminates the training process when the validation performance no longer improves. Different patience values were evaluated to determine whether stopping the training earlier could improve the model’s ability to generalize. Although EarlyStopping reduced the overall training time, it did not consistently improve the prediction accuracy and was therefore not included in the final model.

The influence of the network architecture was also investigated by increasing the number of neurons in the hidden layers from 16 to 32. A larger network provides more trainable parameters and may be able to learn more complex relationships between the extracted physiological features and the target KSS values. However, the experimental results showed that the larger architecture did not consistently outperform the original network. In several experiments, the increase in model complexity produced similar or slightly worse validation performance.

Based on these optimization experiments, the final Regression ANN retained two hidden layers with 16 neurons each. This configuration provided the best balance between prediction accuracy, computational efficiency, and generalization performance, making it the architecture selected for all subsequent evaluations.

III. Results

A. Validation Strategy

To evaluate the generalization capability of the proposed Regression Artificial Neural Network (ANN), three different validation strategies were investigated: Chunk Split, Session Split, and Patient Split. Each strategy represents a different evaluation scenario and measures the model’s ability to generalize under increasingly challenging conditions.

In the Chunk Split, the extracted one-minute data segments were randomly divided into 70% training data and 30% testing data. Although this strategy usually produces the lowest prediction error, samples originating from the same recording may appear in both sets, potentially leading to optimistic performance estimates due to data leakage.

The Session Split separates complete recording sessions. Earlier sessions were used for training, while later sessions were reserved for testing. This evaluates whether the model can successfully predict future recordings from participants that were already observed during training.

Finally, the Patient Split represents the most challenging validation strategy. In this experiment, the model was trained using recordings from four participants and evaluated on recordings from three completely unseen participants. This provides the most realistic estimate of how the model would perform in practical applications where new drivers are encountered.

B. Experimental Results

The prediction performance of the Regression ANN was evaluated using Mean Absolute Error (MAE), where lower values indicate better prediction accuracy. Table I summarizes the obtained results for the three validation strategies.

TABLE I
Regression ANN performance using different validation strategies.

Validation Strategy	MAE
Chunk Split	0.74
Session Split	0.79
Patient Split	1.87

The Chunk Split achieved the lowest prediction error with an MAE of 0.74. While this indicates high prediction accuracy, the result is likely influenced by the random distribution of data chunks, allowing recordings from the same driving session to appear in both the training and testing sets.

The Session Split produced a slightly higher MAE of 0.79. Since complete recording sessions were separated, this experiment better reflects the model’s capability to predict later sessions of participants already encountered during training.

The Patient Split resulted in the highest prediction error, with an MAE of 1.87. Because the testing participants

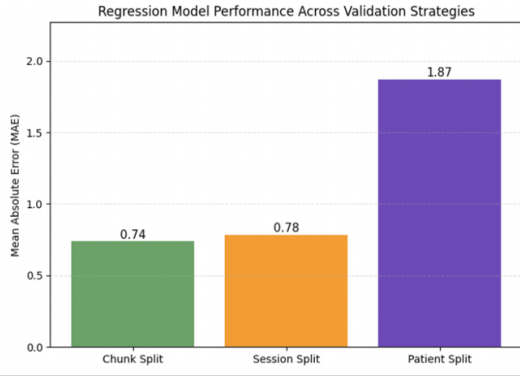


Fig. 3. Mean Absolute Error obtained using the three validation strategies.

were completely excluded from the training process, this experiment represents the most realistic evaluation of the model’s ability to generalize to unseen individuals.

C. Result Interpretation

The obtained results demonstrate that the selected validation strategy has a significant influence on the measured model performance. As the evaluation becomes more realistic, the prediction error increases considerably.

Although the Chunk Split achieved the lowest MAE of 0.74, this strategy may overestimate the true predictive capability of the model because data from the same recording can appear in both the training and testing sets. Consequently, the network may learn recording-specific patterns instead of physiological characteristics that generalize across different situations.

The Session Split produced a slightly higher MAE of 0.79. Since complete recording sessions were separated, this evaluation provides a more reliable estimate of the model’s ability to predict future recordings of participants that were already observed during training. The relatively small increase in error indicates that the proposed Regression ANN is capable of learning stable relationships between the extracted PPG features and the corresponding KSS values.

The Patient Split resulted in the highest prediction error (MAE = 1.87). This experiment represents the most challenging evaluation because the model was tested on participants that were completely excluded from training. The increase in prediction error demonstrates that physiological characteristics vary substantially between individuals, making subject-independent prediction considerably more difficult.

Overall, these findings indicate that the proposed Regression ANN successfully learns meaningful relationships between the extracted PPG features and driver drowsiness. However, the considerable increase in prediction error for the Patient Split highlights that inter-subject variability remains the primary challenge for physiological

machine learning models and should be carefully considered when designing real-world driver monitoring systems.

IV. Discussion

The proposed Regression ANN demonstrated that physiological features extracted from PPG recordings [7] can be used to estimate driver drowsiness with promising prediction accuracy. The experimental results show that the model performs well when evaluated using Chunk and Session Splits but experiences a noticeable decrease in performance under the Patient Split validation strategy.

The differences between the three validation strategies demonstrate that the evaluation methodology has a considerable influence on the measured performance of machine learning models. While random chunk-based splitting produces the lowest prediction error, it may not accurately reflect real-world deployment because recordings from the same participant can appear in both the training and testing sets. In contrast, the Patient Split provides a much more realistic assessment of the model’s ability to generalize to completely new drivers.

One advantage of the proposed approach is that it relies exclusively on PPG-derived features [7]. Compared with camera-based driver monitoring systems, physiological measurements are less affected by environmental conditions such as lighting, camera position, or partial facial occlusions. Furthermore, the compact ANN architecture contains relatively few trainable parameters, resulting in low computational complexity and making the model suitable for real-time implementation in driver monitoring systems.

Another important aspect of this work was the correction of the channel mapping issue encountered during data acquisition. Identifying the correct PPG channel ensured that reliable physiological features could still be extracted despite the recording error. This emphasizes the importance of validating sensor data before feature extraction and model development, since incorrect signal assignments can significantly affect the performance of subsequent machine learning algorithms.

Despite the encouraging results, several limitations remain. The available dataset contains recordings from only a limited number of participants, restricting the model’s ability to learn subject-independent physiological patterns. In addition, the network uses only six handcrafted PPG features. Although these features capture important cardiovascular characteristics, they cannot fully represent the complexity of physiological responses associated with fatigue.

Future work could improve the proposed system by collecting data from a larger and more diverse participant population. Additional physiological signals such as ECG, EDA, or respiration could also be incorporated to provide complementary information about driver fatigue. Furthermore, multimodal approaches that combine physiological features with facial measurements such as Eye

Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) may improve prediction accuracy and increase robustness under real-world driving conditions. More advanced deep learning architectures could also be investigated to further enhance generalization performance while maintaining computational efficiency.

V. Conclusion

This work presented a Regression Artificial Neural Network for predicting driver drowsiness using six physiological features extracted from PPG recordings. The proposed model successfully estimated continuous Karolinska Sleepiness Scale (KSS) values and was evaluated using three different validation strategies.

The experimental results showed that the Chunk Split achieved the lowest prediction error, while the Patient Split provided the most realistic evaluation of the model's generalization capability. These findings demonstrate that validation strategy has a significant impact on the measured performance of physiological machine learning models.

Overall, the developed Regression ANN provides a promising approach for non-invasive driver drowsiness prediction. The obtained results indicate that meaningful relationships can be learned from a compact set of PPG-derived features while maintaining a relatively simple network architecture. Although the proposed model achieved encouraging performance, the Patient Split results show that improving subject-independent generalization remains an important challenge for future research.

Future work should focus on collecting data from a larger and more diverse participant population, integrating additional physiological signals, and investigating more advanced neural network architectures. Combining physiological measurements with visual fatigue indicators may further improve prediction accuracy and increase the robustness of driver monitoring systems in real-world applications.

References

- [1] A. Kecklund and T. Åkerstedt, "The Karolinska Sleepiness Scale: Correlation with EEG and performance," Department of Clinical Neuroscience, Karolinska Institute, Stockholm, Sweden, 1992.
- [2] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," Proc. 3rd Int. Conf. Learning Representations (ICLR), San Diego, CA, USA, 2015.
- [3] V. Nair and G. E. Hinton, "Rectified Linear Units Improve Restricted Boltzmann Machines," Proc. 27th Int. Conf. Machine Learning (ICML), Haifa, Israel, pp. 807–814, 2010.
- [4] M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," Software available: <https://www.tensorflow.org>, 2016.
- [5] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.
- [6] H. Silva et al., "BITalino: A Biosignal Acquisition System Based on the Arduino," Proc. Int. Conf. Biomedical Electronics and Devices (BIODEVICES), 2014.
- [7] J. Allen, "Photoplethysmography and its Application in Clinical Physiological Measurement," Physiological Measurement, vol. 28, no. 3, pp. R1–R39, 2007.

- [8] S. Kaplan, M. A. Guvensan, A. G. Yavuz, and Y. Karalurt, "Driver Behavior Analysis for Safe Driving: A Survey," IEEE Transactions on Intelligent Transportation Systems, vol. 16, no. 6, pp. 3017–3032, Dec. 2015.